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# ReAct Prompting | Prompt Engineering Guide

7-9 minutes

Yao et al., 2022 (opens in a new tab) introduced a framework named ReAct where LLMs are used to generate both *reasoning traces* and *task-specific actions* in an interleaved manner.

Generating reasoning traces allow the model to induce, track, and update action plans, and even handle exceptions. The action step allows to interface with and gather information from external sources such as knowledge bases or environments.

The ReAct framework can allow LLMs to interact with external tools to retrieve additional information that leads to more reliable and factual responses.

Results show that ReAct can outperform several state-of-the-art baselines on language and decision-making tasks. ReAct also leads to improved human interpretability and trustworthiness of LLMs. Overall, the authors found that best approach uses ReAct combined with chain-of-thought (CoT) that allows use of both internal knowledge and external information obtained during reasoning.

## How it Works?

ReAct is inspired by the synergies between "acting" and "reasoning" which allow humans to learn new tasks and make decisions or reasoning.

Chain-of-thought (CoT) prompting has shown the capabilities of LLMs to carry out reasoning traces to generate answers to questions involving arithmetic and commonsense reasoning, among other tasks (Wei et al., 2022) (opens in a new tab). But its lack of access to the external world or inability to update its knowledge can lead to issues like fact hallucination and error propagation.

ReAct is a general paradigm that combines reasoning and acting with LLMs. ReAct prompts LLMs to generate verbal reasoning traces and actions for a task. This allows the system to perform dynamic reasoning to create, maintain, and adjust plans for acting while also enabling interaction to external environments (e.g., Wikipedia) to incorporate additional information into the reasoning. The figure below shows an example of ReAct and the different steps involved to perform question answering.

